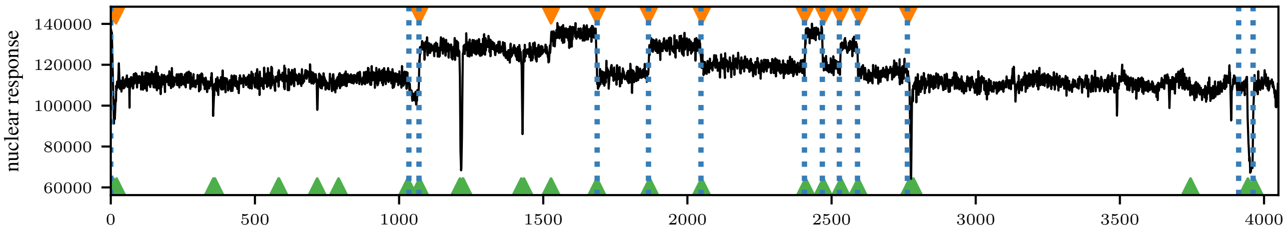

Figure 3. Well-log data. MAP segmentation indicated by **blue** dashed lines for $\mathcal{D}_m$-BOCD, ▼ for β-BOCD, and ▲ for standard BOCD. Standard BOCD mistakenly labels outliers as CPs, while both $\mathcal{D}_m$-BOCD and β-BOCD are robust and identify lasting changes.

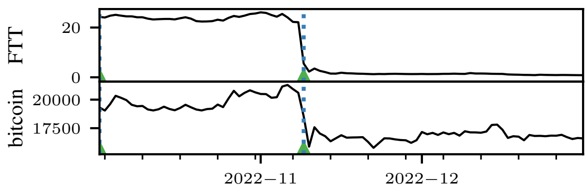

Figure 4. Crypto-crash. MAP segmentation indicated by **blue** dashed lines for $\mathcal{D}_m$-BOCD, and by ▲ for standard BOCD. There are no outliers, so both methods identify the correct CP.

known issue with robust CP detection method is that they can experience a latency when it comes to detecting actual CPs. Interestingly, this is not something we observe in our experiments with this choice of m and θ^*.

Choice of ω How to choose ω is an important question for generalised Bayesian inference, and has more than one answer (Lyddon et al., 2019; Syring & Martin, 2019; Matsubara et al., 2022a; Bochkina, 2022; Wu & Martin, 2023). Previous methods are computationally expensive, asymptotically motivated, and focus on tuning the learning rate to provide asymptotically correct frequentist coverage. As the computational overhead of these methods is substantial and their asymptotic arguments generally do not apply to the CP setting, we pursue a different strategy: we match the uncertainty of the generalised posterior to that of its standard counterpart on the first $t^\star$ observations of the data stream. To operationalise this, we choose

$$\omega^\star = \arg\min_{\omega>0} \mathrm{KL}\left(\pi_\omega^{\mathcal{D}_m}(\theta|x_{1:t^\star}) \| \pi^{\mathrm{B}}(\theta|x_{1:t^\star})\right).$$

Computing $\omega^\star$ is implemented using automatic differentiation via `jax` (Bradbury et al., 2018). This is possible even if the standard Bayes posterior π^{B} is intractable, since $\pi_\omega^{\mathcal{D}_m}$ has a conjugacy property (see Proposition 3.1). Since the standard Bayes posterior is reliable in the absence of outliers and heterogeneity, this yields reasonable uncertainty quantification if the degree of misspecification is mild at the beginning of the data stream. Our experiments confirm this: the uncertainty is well-calibrated, both predictively and with regards to the run-length posterior.

4. Experiments

We investigate $\mathcal{D}_m$-BOCD empirically in several numerical experiments. In doing so, we highlight its computational and inferential advantages over standard BOCD and β-BOCD. In all experiments, we choose conjugate priors as in Proposition 3.1, and m and ω as in Section 3.4. All code and data is publicly available at `https://github.com/maltamiranomontero/DSM-bocd`.

Computational complexity. We compare the complexity of the three BOCD methods in different settings and show that $\mathcal{D}_m$-BOCD is considerably faster than β-BOCD, even when sampling is needed. Moreover, Figure 5 shows that $\mathcal{D}_m$ is as fast as standard BOCD when $d = 1$ and the predictive posterior is available in closed form. See Appendix C.1 for details.

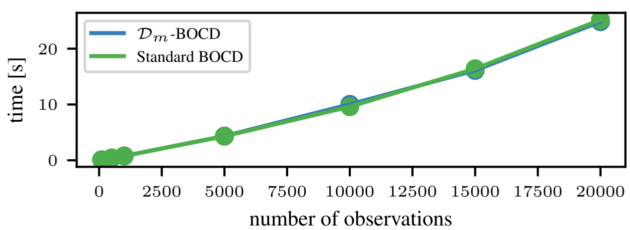

Figure 5. Overall time in seconds versus number of observations. We observe that both methods are equally fast for any number of observations.

Accuracy and detection delay. We quantify the method's performance advantage by comparing detection delay and accuracy on artificially generated data with outliers. We generate 600 samples with 2% of outliers and 6 CPs; then, we report the positive predictive value (PPV), true positive rate (TPR), and detection delay. See Appendix C.3 for the exact expression of the metrics. Table 1 shows that our method detects the same amount of true positives as the standard BOCD while not detecting many false positives, showing the strength of our method. Moreover, the detection delay shows that in spite of being robust to outliers, $\mathcal{D}_m$-BOCD does not cause any delay in the detection of CP.

Method	PPV	TPR	Delays
$\mathcal{D}_m$-BOCD	**0.907±0.154**	**0.883±0.13**	1.643±0.475
Standard BOCD	0.6±0.128	0.833±0.149	**1.05±1.545**

Table 1. Performance indices-mean and standard deviation-using the positive predictive value (PPV), true positive rate (TPR), and the detection delays over 10 realisations. For PPV and TPR, the nearest to 1, the better. For delays, the lower, the better.

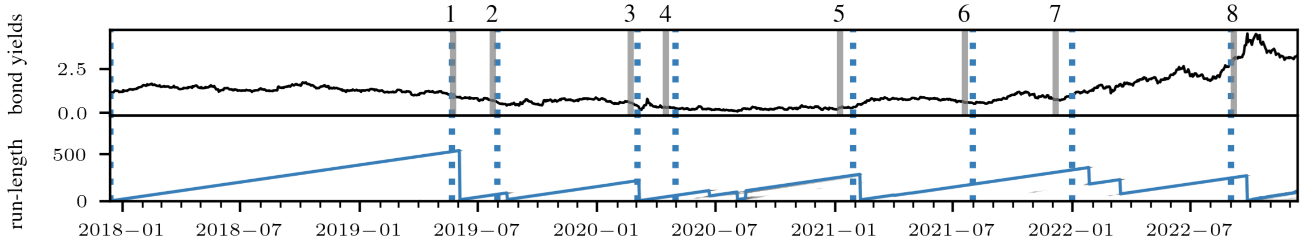

Figure 6. UK's 10 year government bond yield 2018-2023. The MAP-segmentation resulting from $\mathcal{D}_m$-BOCD is indicated in dashed **blue** lines. The bottom panel displays the corresponding run-length posterior, with the most likely run-length marked in **blue**. A series of political events of national importance closely track the segmentation, and are marked with solid **gray** lines: 1. Theresa May announces her resignation from her position as prime minister; 2. Boris Johnson sworn in as prime minister; 3. the first Covid case recorded in EU; 4. the first Covid wave in the UK is officially declared; 5. the third Covid wave in the UK is officially declared; 6. the legal limits on social contact removed in UK; 7. Covid 'Plan B' measures are implemented in UK in response to the spread of the Omicron variant; 8. Liz Truss is sworn in as prime minister.

Twitter flash crash & Cryptocrash. A robust CP detection algorithm must not to be fooled by outliers while detecting CP correctly. We show that $\mathcal{D}_m$-BOCD has this capability on two real-world examples: the first is the Dow Jones Industrial Average (DJIA) index every minute on 17/04/2013, the day of the *Twitter flash crash*. The data is publicly available on FirstRate Data.[1] That day, the Associated Press' Twitter account was hacked and falsely tweeted that explosions at the White House had injured then-president Barack Obama. In response, the DJIA dropped by 150 points in a matter of seconds before bouncing back. As Figure 1 shows, this is a clear outlier. Modelling the time series with a Gaussian, the plot shows that $\mathcal{D}_m$-BOCD successfully ignores this blip, while standard BOCD incorrectly labels it as a CP. The second example tracks the average daily value of FTT and Bitcoin between 10/2022 and 12/2022, data which is publicly available on Yahoo finance.[2] FTT was the token issued by FTX, one of the biggest crypto-exchanges before it failed due to a liquidity crisis on November 11th 2022. The ensuing collapse of FTX marked a crash in the value of various crypto-currencies, including Bitcoin. Using a two-dimensional Gaussian distribution for both $\mathcal{D}_m$-BOCD and standard BOCD, Figure 4 shows that both methods correctly detect the CP. Figure 12 in Appendix C also displays the run-length posteriors, and shows that robustness does not lead to increased CP detection latency.

Well-log. The well-log data was introduced in Ruanaidh & Fitzgerald (1996), and consists in 4,050 nuclear magnetic resonance measurements recorded while drilling a well. CPs in the sequence correspond to changes in the sediment layers the drill is penetrating. On top of these clear changes, the data contains outliers and contaminants corresponding to more short-term events in geological history—such as flooding, earthquakes, or volcanic activity. When this data set is studied, its outliers have traditionally been removed before CP detection algorithms are run (see e.g. Adams & MacKay, 2007; Ruggieri & Antonellis, 2016; Levy-leduc & Harchaoui, 2008). We leave them in, and Figure 3 shows that this is unproblematic for $\mathcal{D}_m$-BOCD, but does lead to falsely labelled CPs with BOCD. We also compare the algorithm with β-BOCD (Knoblauch et al., 2018), and find that the detected changes are almost identical. On a machine with processor Intel i7-7500U 2.7 GHz, and 12GB of RAM, $\mathcal{D}_m$-BOCD took about 10 times less than β-BOCD.

Multivariate synthetic data. In certain settings, $\mathcal{D}_m$-posteriors are conjugate when standard posteriors are not. An example is a multivariate time series whose dimensions follow different distributions belonging to the exponential family. To this end, we generate 1000 samples from a time series with CPs at $t = 250, 750$. Conditional on the CPs, the data is generated independently from an exponential in the first dimension and Gaussian distribution in the second dimension. $\mathcal{D}_m$-BOCD is immediately applicable, and Figure 7 shows that the algorithm functions reliably. We do not compare to BOCD in this setting: for this model, standard Bayesian posteriors would require expensive sampling algorithms or variational approximations to be employed, rendering the algorithm impractical.

UK 10 year government bond yield. Finally, we run the $\mathcal{D}_m$-BOCD on the daily yield of 10 year UK government bonds from 2018 to 2022 (see Figure 6). The data is publicly available via the Bank of England database.[3] Since the 10-year yield has been positive throughout history, we model it using the gamma distribution. As shown in Figure 6, we detect changes in the yield curve that correspond to important political events in the UK. This distribution leads to a $\mathcal{D}_m$-posterior that is a Gaussian truncated at zero. For standard Bayes, a conjugate prior exists, but it leads to a posterior with intractable normalisation constant. Like the multivariate synthetic data example, this constitutes another instance where $\mathcal{D}_m$-posteriors have better computational properties than standard Bayes.

[1] https://firstratedata.com/free-intraday-data

[2] https://finance.yahoo.com/

[3] https://www.bankofengland.co.uk/boeapps/database/

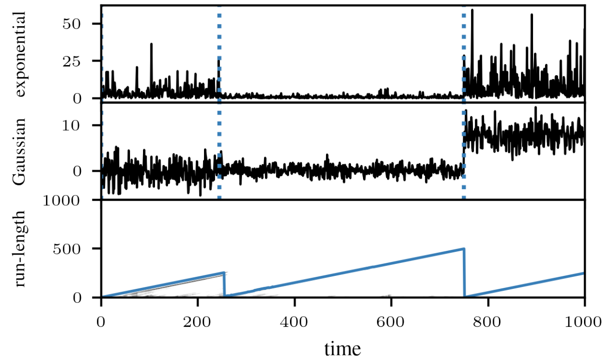

Figure 7. Multivariate synthetic example. A 2-dimensional CP problem. For the chosen model, $\mathcal{D}_m$-BOCD is computationally efficient, but standard BOCD is computationally infeasible. The MAP segmentation is indicated by dashed **blue** lines, and the bottom panel shows the run-length distribution, with the most likely value in **blue**.

5. Conclusion

We proposed $\mathcal{D}_m$-BOCD, a new version of BOCD that is both *robust to outliers and scalable*. The algorithm relies on a new generalised Bayesian inference scheme constructed with diffusion score-matching. These posteriors have closed form updates for models that are members of the exponential family, and provide robustness by appropriately tuning the diffusion matrix m. For T observations, d-dimensional data, and p model parameters, the overall run time of the method is $\mathcal{O}(T(p^2 + d^2))$, and we demonstrate that it is just as fast as standard BOCD. By showcasing the various computational and inferential benefits of $\mathcal{D}_m$-BOCD on a range of examples, we demonstrate that it is a powerful and needed addition to the literature. In the future, we will also investigate the applicability of $\mathcal{D}_m$-BOCD to regression models. This is not trivial: the regression setting changes both the definition of valid score matching losses, as well as how to show their robustness (Xu et al., 2022).

$\mathcal{D}_m$-posteriors also are of independent interest for computational challenges in Bayesian inference: like the generalised posterior in Matsubara et al. (2022b) and Matsubara et al. (2022a), they can be computed even without access to the normalising constant of the likelihood. This suggests that $\mathcal{D}_m$-posteriors should be studied more broadly as a potential competitor to other Bayesian methods for intractable likelihood problems.

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References

Adams, R. P. and MacKay, D. J. Bayesian online changepoint detection. *arXiv:0710.3742*, 2007.

Agudelo-España, D., Gomez-Gonzalez, S., Bauer, S., Schölkopf, B., and Peters, J. Bayesian online prediction of change points. In *Conference on Uncertainty in Artificial Intelligence*, pp. 320–329, 2020.

Anastasiou, A., Barp, A., Briol, F.-X., Ebner, B., Gaunt, R. E., Ghaderinezhad, F., Gorham, J., Gretton, A., Ley, C., Liu, Q., Mackey, L., Oates, C. J., Reinert, G., and Swan, Y. Stein's method meets computational statistics: A review of some recent developments. *Statistical Science*, 38: 120–139, 2023.

Barp, A., Briol, F.-X., Duncan, A., Girolami, M., and Mackey, L. Minimum Stein discrepancy estimators. In *Advances in Neural Information Processing Systems*, pp. 12964–12976, 2019.

Barry, D. and Hartigan, J. A. A Bayesian analysis for change point problems. *Journal of the American Statistical Association*, 88(421):309–319, 1993.

Basu, A., Harris, I. R., Hjort, N. L., and Jones, M. Robust and efficient estimation by minimising a density power divergence. *Biometrika*, 85(3):549–559, 1998.

Bissiri, P. G., Holmes, C. C., and Walker, S. G. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):1103–1130, 2016.

Bochkina, N. Bernstein-von Mises theorem and misspecified models: a review. *arXiv:2204.13614*, 2022.

Boustati, A., Akyildiz, O. D., Damoulas, T., and Johansen, A. Generalised Bayesian filtering via sequential Monte Carlo. In *Advances in Neural Information Processing Systems*, pp. 418–429, 2020.

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., and Zhang, Q. JAX: composable transformations of Python+NumPy programs, 2018. URL `http://github.com/google/jax`.

Chérief-Abdellatif, B.-E. and Alquier, P. MMD-Bayes: Robust Bayesian estimation via maximum mean discrepancy. In *Symposium on Advances in Approximate Bayesian Inference*, 2020.

Chernozhukov, V. and Hong, H. An MCMC approach to classical estimation. *Journal of Econometrics*, 115(2): 293–346, 2003.